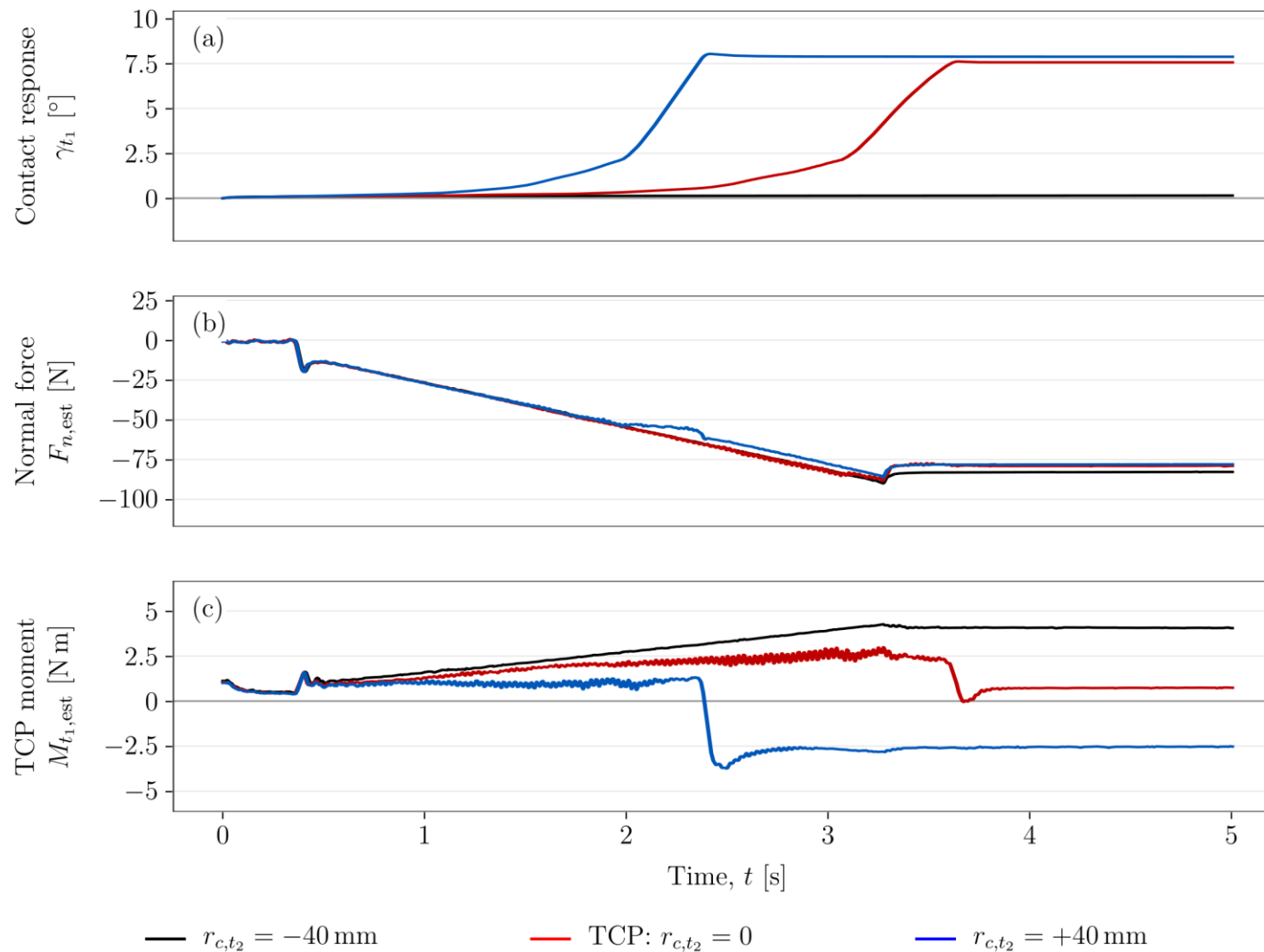


Contact response, normal force and moment



Representative traces Positive entry tilt

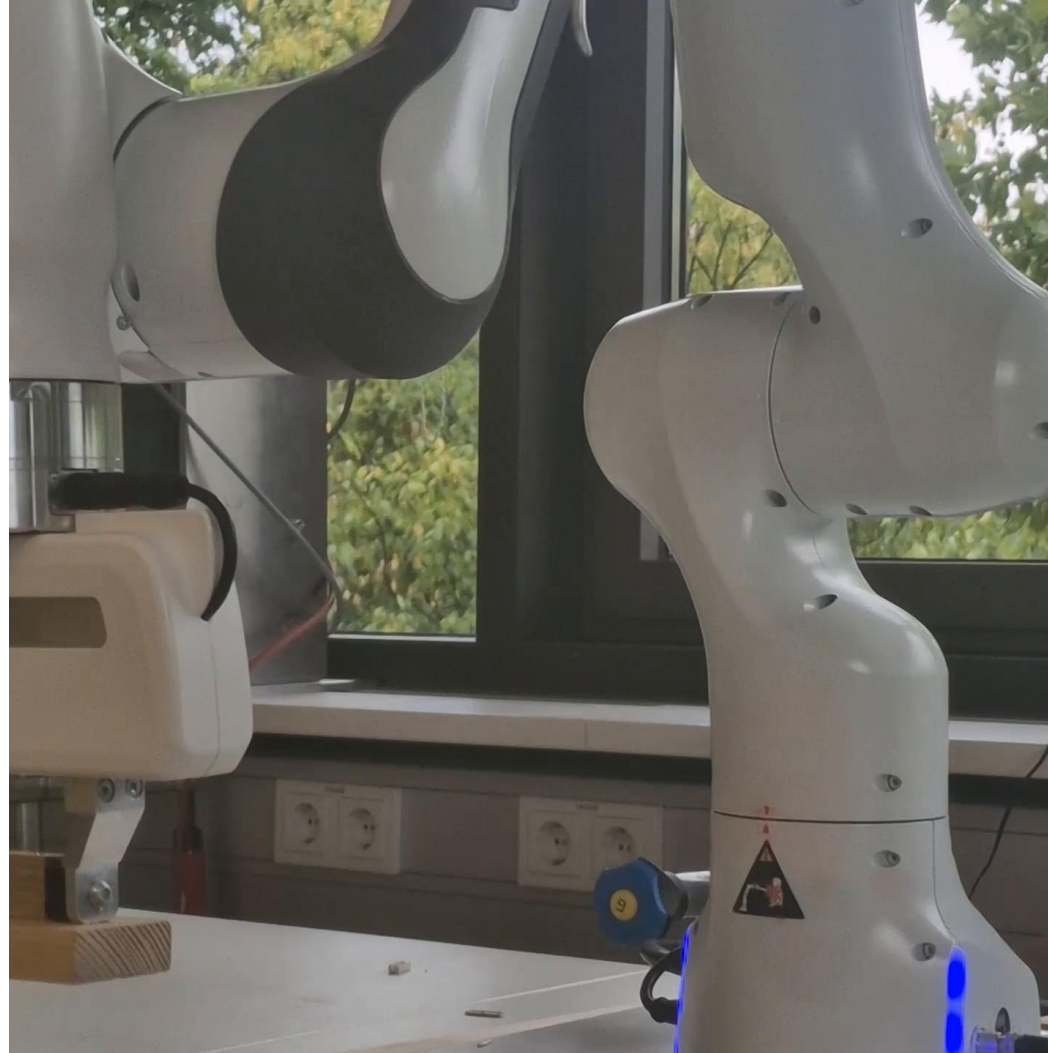
- Similar final normal force
- Moment reverses with CoC displacement
- Force and moment: model-based robot estimates

Contact demonstration



Pre-grinding hold · fixed desired orientation · CoC at TCP

Disturbance demonstration



No null-space torque → Damping → Conditioning